🏛️ SkillBridge — Master Architectural Blueprint & Roadmap

> **Tagline:** "*Share. Learn. Repeat.*"
> **Mission:** Community-driven, zero-mandatory-cost peer-to-peer knowledge exchange for college students.
> **Core Concept:** "*I know something you want to learn, and you know something I want to learn.*"

1. System Philosophy & Anti-AI Realism Guardrails

SkillBridge is built as a Community Engagement Project (CEP) engineered to production-grade SaaS standards. To prevent the "cheap generic template" look, every interface, dataset, and interaction is bound to these standards:

- **Persona Realism:** Indian engineering campus demographics (e.g., Aarav Sharma from IIT Bombay, Priya Patel from VJTI, Rohan Verma from BITS Pilani). Realistic tech-stack pairings (e.g., teaching *React & Tailwind* while learning *Data Structures & Algorithms in C++* or *System Design*).
- **Design Tokens:**
  - **Background / Canvas:** Warm Cream (`#FDFBF7`)
  - **Surfaces / Cards:** Pure Crisp White (`#FFFFFF`)
  - **Brand Primary:** Deep Burgundy / Wine (`#722F37`), hover: `#5A252C`
  - **Accents & Badges:** Soft Peach (`#FFB085`) and Light Almond (`#FFE8D6`)
  - **Typography:** Plus Jakarta Sans & Inter
  - **Headings & Text:** Dark Slate (`#2D3748`) and Muted Grey (`#718096`)
- **Micro-Interactions & Structure:**
  - Glassmorphic navigation headers (`backdrop-blur-md bg-[#FDFBF7]/80`)
  - Bento-grid responsive dashboard modules
  - Diffused multi-layer drop shadows (`shadow-[0_8px_30px_rgb(0,0,0,0.04)]`)
  - Organic rounded radii (`rounded-2xl`, `rounded-3xl`)
  - Shimmering skeleton loaders (no raw spinning spinners)
  - Calculated algorithmic compatibility badges (e.g., `94% Match`, `81% Match`)

2. Part A: Complete Application Architecture & User Flow

2.1 High-Level Architecture Diagram

2.2 End-to-End User Journeys

Journey 1: New Student Onboarding & Matching

Journey 2: Skill Exchange Request to Learning Room

3. Part B: Database ER Diagram & Schema Specifications

3.1 Relational Entity-Relationship Diagram

4. Part C: Recommended Tech Stack for a 4-Person Student Team

4.1 Team-Friendly, Zero-Headache Stack Matrix

Layer	Recommended Choice	Rationale for a 4-Person Student Team
Framework	**Next.js 15 (App Router)**	Full-stack in one repo. Frontend UI and backend API routes in TypeScript eliminate cross-repo synchronization friction.
Language	**TypeScript 5**	Strict type definitions prevent runtime bugs across teammates. Autocomplete speeds up development for beginners.
Styling & UI	**Tailwind CSS + Lucide Icons + Tailwind Variants**	Rapid styling with custom palette tokens (`#722F37` , `#FDFBF7` , `#FFB085`). Zero heavy CSS files to maintain.
Database & ORM	**Prisma ORM with SQLite (Local Dev) → PostgreSQL (Supabase/Neon)**	Team can clone and run `npx prisma db push` instantly on local SQLite without needing a heavy local Docker or Postgres setup on day one, then switch 1 line of config to deploy to Neon/Supabase for production.
Authentication	**NextAuth.js (Auth.js) / Secure Session Cookies**	Industry standard, zero ongoing SaaS fee, supports credentials and Google/GitHub OAuth easily.
State & Fetching	**React Hooks + SWR / Server Actions**	Minimal boilerplate, automatic optimistic UI updates, zero Redux bloat.
Media / Virtual Room	**WebRTC Mock UI + Live Timer & Sync Notes Engine**	Interactive meeting suite with real webcam/mic toggling, screen-share preview, live dual-pane collaborative notes, and countdown timers.
Deployment	**Vercel + Neon PostgreSQL / Supabase**	100% free-tier eligible for college project presentations; automated CI/CD directly from GitHub on every pull request.

4.2 4-Person Workload Division

- **Developer 1 (Architect / Full-Stack Lead):** Database schema, NextAuth configuration, Matching Algorithm logic, API Route handlers.
- **Developer 2 (UI/UX & Design System Lead):** Global layout, warm-cream/burgundy theme tokens, Public Pages (Home, About, Explore), Bento Dashboard components.
- **Developer 3 (Core Interaction Lead):** Profile Onboarding wizard, Exchange Request flow, Private Chat messaging, and Session Scheduling modals.

- **Developer 4 (Features & Engagement Lead):** Interactive Learning Room (Timer, Notes, Mock Controls), Post-Session Feedback & Points Ledger, Community Doubt Forum, and Admin Panel.

5. Part D: 6-Factor Matching Algorithm Mathematical Specification

The match score between User AA (learner) and User BB (potential mentor) is computed on a scale of 00 to $100$100:

$$\text{Total Match Score} = S_{\text{skill}} + A_{\text{avail}} + L_{\text{lang}} + C_{\text{level}} + M_{\text{mode}} + R_{\text{points}}$$

Total Match Score = Sskill + Aavail + Llang + Clevel + Mmode + Rpoints

Where:

1. **Skill Compatibility ($S_{\text{skill}} \in [0, 50]$Sskill ∈ [0, 50] pts):**
    - Direct mutual match (AA wants what BB teaches AND BB wants what AA teaches): 5050 pts.
    - One-way match (BB teaches what AA wants): 3535 pts.
    - Related category match: 2020 pts.
2. **Availability Overlap ($A_{\text{avail}} \in [0, 15]$Aavail ∈ [0, 15] pts):**
    - Shared free day/time slot: 1515 pts (55 pts per overlapping window up to 1515).
3. **Language Match ($L_{\text{lang}} \in [0, 10]$Llang ∈ [0, 10] pts):**
    - At least 1 shared spoken language: 1010 pts.
4. **Level Compatibility ($C_{\text{level}} \in [0, 10]$Clevel ∈ [0, 10] pts):**
    - Mentor level is strictly greater than or equal to learner's current target level: 1010 pts; otherwise 55 pts.
5. **Mode / Location Preference ($M_{\text{mode}} \in [0, 10]$Mmode ∈ [0, 10] pts):**
    - Exact mode match (Online-Online, or same campus offline): 1010 pts; Hybrid: 77 pts; Mismatch: 00 pts.
6. **Reputation & Points Momentum ($R_{\text{points}} \in [0, 5]$Rpoints ∈ [0, 5] pts):**
    - High rating ($\geq 4.5/5$≥ 4.5/5) and active peer reviews: 55 pts.

6. Part E: Master 17-Phase Development Roadmap

Phase	Milestone	Core Deliverables
Phase 1	**Repository Foundation & Design System**	Next.js 15 + Tailwind CSS setup, custom color palette (`#FDFBF7` , `#722F37` , `#FFB085`), typography, Lucide icons, button variants, and base Card components.
Phase 2	**Prisma Database Schema & Seed Data**	Comprehensive Prisma schema (12 tables), SQLite/PostgreSQL setup, realistic seed dataset with 10 authentic Indian engineering student profiles and diverse skill portfolios.
Phase 3	**Public Landing Page & Navigation**	High-converting Hero section with dynamic floating cards, "How it Works" 3-step carousel, live community stats, student testimonials, and responsive glassmorphic navbar/footer.
Phase 4	**About Us & Philosophy Page**	Story of SkillBridge, peer-to-peer manifest, "Zero Mandatory Money" ethos, interactive comparison table (SkillBridge vs paid EdTech).
Phase 5	**Authentication System**	Clean modal-driven Sign-In and Sign-Up flows, college email validation, password reset mockup, and persistent mock session provider.
Phase 6	**Multi-Step Profile Onboarding Flow**	4-step progressive wizard (Skills to Teach, Skills to Learn, Availability Grid, Preferred Mode & Languages) with instant validation and completion progress bar.
Phase 7	**Bento-Grid Student Dashboard**	Dashboard hub: Recommended peer matches, upcoming session countdown, pending incoming/outgoing requests, Skill Points balance widget, and live notification drawer.
Phase 8	**Explore Skills & Directory Engine**	Full search catalog, filter by category (AI, Web, Design, DSA, Core Eng), filter by proficiency level, sort by rating/points, and mentor detail previews.
Phase 9	**Algorithmic "My Matches" Hub**	Dedicated matching screen showcasing calculated match percentages (84%84%, 92%92%), breakdown tooltip (Skills, Availability, Language), and instant "Request Exchange" CTA modal.
Phase 10	**Real-Time Direct Chat & File Sharing**	WhatsApp/Slack-grade dual-pane chat interface, peer list, message bubbles, active status, mock attachment uploader, and quick "Propose Session" button.
Phase 11	**Exchange Request & Scheduling Flow**	Date/time picker modal, skill selection dropdown, invitation state management (Pending → → Accepted → → Scheduled), and calendar integration.
Phase 12	**Interactive Learning Room Suite**	Dedicated room view with mock video/audio stream toggles, screen-share preview, real-time 60-minute countdown timer, dual-pane markdown notes editor, and in-room chat sidebar.

Phase	Milestone	Core Deliverables
Phase 13	**Post-Session Review & Skill Points Engine**	5-star rating modal, tags for peer feedback ("Patient", "Great Visuals", "Clear Explanations"), automatic +25 points credit, and points ledger history.
Phase 14	**Community Doubt & Problem Board**	StackOverflow-inspired college doubt forum: post a question, filter by tags (Recursion, Next.js, Dynamic Programming), upvote answers, mark verified solution, and earn karma.
Phase 15	**Comprehensive User Profile**	Public/Private profile view: skills taught/learning badges, student bio, college badge, verified session tally, points ranking, and peer reviews list.
Phase 16	**Settings & Security Suite**	Account management, privacy toggles (hide profile from public search), notification preferences (email/in-app), password change, and session logout.
Phase 17	**Admin Control & Moderation Suite**	Executive dashboard: total users, active exchanges, platform health metrics, skill taxonomy manager, and user moderation/report resolution panel.

7. Approval Checkpoint

This completes the comprehensive blueprint for **SkillBridge**.
Upon your explicit approval:

1. We will begin directly with **Phase 1 (Repository Foundation & Design Tokens)**.
2. Every file will be built cleanly with its full path, complete code, exact terminal installation commands, and step-by-step verification instructions.